

Environmental Impact of Ai

Posted by u/Glass-Ad672 • 0 points • 18 comments

So what is the actual environment impact of ai data centers exactly? Some people say it takes 5 kagillions gallons of water, other say it's not even a factor. So what is the actual water usage and any other environmental impacts of note? Also, please provide a source so I can verify what's being said. As much as I would love to take your word as truth, I simply cant.

Comments

u/FactsOverFeelings-69 • 1 points • 2026-01-21 14:52 UTC

Most of the confusion is just mixed context. Training, daily usage, cooling method, and location all change the numbers a lot. Looking at cited sources side by side usually clears it up. I sometimes use AlchemistAiOfficial for quick checks because it shows sources instead of raw claims.

u/Human_certified • 1 points • 2026-01-16 16:30 UTC

AI is an industry, not a tiny hobby. At the global level, small amounts add up. But at the global level, *everything* adds up.

By late 2025, best estimates put AI at around <0.05% of global electricity (not energy) use. For something with 1.6 billion users, that's about as efficient as you can get.

u/Tarc_Axiom • 1 points • 2026-01-16 15:38 UTC

First, you need to understand what data centres are.

People keep acting like machine learning has created the concept of data centres, which is obviously not true. Data centres as massive projects with enormous infrastructure demands had existed for decades before machine learning had any kind of public/commercial presence. Furthermore, machine learning as a subset of computing represents about 20% of the global data centre infrastructure, today. Though that number will obviously rise, and that's important because...

As a society, we use the resources available to us to do things. This is a concept that a lot of people who make this argument do not seem to understand. Yes, data centres use electricity and water (though, water in extremely small amounts). You know what else uses electricity and VASTLY more water? Agriculture. Farming. **Food**. Why aren't these people so aggressively against the water cost of creating food? If I had to guess, it's because we need food, **and as a modern society, we need technology too.**

Someone will undoubtedly reply to this comment with a big longwinded reply about why we don't need

machine learning and that it is all bad for society, and it is your job to ignore that person because they don't know what they're talking about and are arguing in bad faith. We need technology, we need the advancements these technologies provide, and we will continue to need ever increasing amounts of this technology for generations. The good news, is that the overall cost of these technologies ****IS NOT**** outpacing their returned value to society, so we're fine.

In regards to your request, there is no accurate source as to the water consumption of machine learning data centres, because water is not consumed and data centres don't actually replace the water they use all the time. Further, the models are more complex and convoluted (in a technical sense) than they have been at any point in the past and they will be more complex and convoluted than they are right now at any point in the future. There are a million other things that make any estimate about the water consumption of machine learning data centres substantively incorrect. You can only look at effects, not estimations, and there aren't many effects to look at.

Sometimes data centres are built in areas where they put strain on a community, that's bad. That however is a failure of corporate greed and capitalism, not a failure of the existence of an advanced technology.

u/Extra_Island7890 • 1 points • 2026-01-16 07:27 UTC

Regardless of how high or low it is, the fact that people think it is high is a great opportunity to get them behind a slowly-rising carbon tax with the revenues distributed as the start of a UBI, and/or carbon reduction efforts like planting trees, or some of these new methods to remove carbon from the air with technology. Joke's on them, it will impact gasoline prices a lot more than Ai - but pollution is not specific to one industry, it needs to be taxed at the source wherever it happens and the tax needs to be proportional to the pollution caused.

u/Enough-Impression-50 • 2 points • 2026-01-16 06:34 UTC

But would YOU jump off this 1000 foot cliff as the first of several challenges across several weeks against 21 other campers for the chance to win 100,000 dollars?

[https://preview.redd.it/nggqqup4pndg1.jpeg?](https://preview.redd.it/nggqqup4pndg1.jpeg?width=566&format=pjpg&auto=webp&s=dd51c0eb036315cd8ba19e5d4f94edcdf75a8906)

[width=566&format=pjpg&auto=webp&s=dd51c0eb036315cd8ba19e5d4f94edcdf75a8906](https://preview.redd.it/nggqqup4pndg1.jpeg?width=566&format=pjpg&auto=webp&s=dd51c0eb036315cd8ba19e5d4f94edcdf75a8906)

u/Glass-Ad672 • 1 points • 2026-01-16 07:02 UTC

hell yeah

u/Dr-False • 2 points • 2026-01-16 05:52 UTC

Might be 300,000 gallons, might be a lot less. What i do know is some datacenters they use can run at 115C which does mean superheated water, though do I know what pressure they keep it at? (more pressure = higher boiling point) not really.

It's one of the reasons I'm not hostile, but critical of AI companies. I really want to see them disclose the numbers mainly so efficiency goals can be achieved. At the end of the day, I don't actually hate the technology, just the lack of transparency and Microsoft shoving Co-Pilot down my throat

u/CuirPig • 3 points • 2026-01-16 05:50 UTC

I hate to say this, but you could get a comprehensive overview that was excellently written with links to references in a matter of seconds if you asked ChatGPT or Gemini. They will tell you not only what the facts are, but how they are being misrepresented by both sides. And because you asked for links, any hallucinations would be obvious. They could also tell you about how the companies are approaching

these new data centers and what they are doing to prevent negative impacts both environmental and economic.

u/Glass-Ad672 • 2 points • 2026-01-16 05:52 UTC

that is kinda funny "how much water does ai use?" "just ask ai", seriously tho, i probably will end up doing that, thanks for the idea

u/JasonP27 • 1 points • 2026-01-16 07:19 UTC

I saw one posted a few days back I took a couple screenshots. I can't vouch for the accuracy but it claims that when comparing water usage you could generate 11,000 AI images per single sheet of paper produced.

<https://preview.redd.it/m5al8k46xndg1.jpeg?width=1079&format=pjpg&auto=webp&s=304572383ecbacd719acf24501a5bae30deb204e>

u/memequeendoreen • 3 points • 2026-01-16 05:46 UTC

I'm not too concerned with the resource usage on its own. I'm concerned with the fact that a data center in your area will drive up energy prices across the board. For what, exactly? Glad you asked. These higher prices will be for students cheating, people online creating illegal content, ripping off artists and offloading the general public's critical thinking skills to some LLM.

I know there are good things that AI does, but it really doesn't do enough good to make me actually think it's worth it.

u/Cable_Typical • 2 points • 2026-01-16 05:31 UTC

A better way to picture it is to compare infrastructure costs of the related things that use datacenters, it'd be a better approximation than spitballing big numbers for the purpose of framing.

u/Candid-Station-1235 • 1 points • 2026-01-16 05:29 UTC

patch your own ignorance

u/MysteriousPepper8908 • 13 points • 2026-01-16 05:29 UTC

Can't be bothered to dig up sources right now but the answer is both. It does use a ton of water, it just turns out if you look at figures in context, a ton isn't that much compared to how much everyone else is using. This is why it's important to understand the scale of water use and orders of magnitude. It's not a primary source but you can just watch Hank Green's video, he's a reputable science educator and far from an AI apologist. [https://www.youtube.com/watch?v=H_c6MWk7PQc]
(https://www.youtube.com/watch?v=H_c6MWk7PQc)

u/Vilefighter • 3 points • 2026-01-16 05:40 UTC

Came here to comment the same thing: watch Hank Green's vid. The answer is complicated as things usually are. My main takeaway personally is that even in the worst case it's a minor environmental impact in the macroscopic sense when compared to other sources of electricity and water consumption, but misuse of resources on a local scale by data centers can be a big problem. But it's possible to watch that video and come away with something different - it feels well balanced in that respect.

u/nuker0S • 2 points • 2026-01-16 12:59 UTC

The takeaway is that you should bully datacenters build in wrong places, not users, or even idea as a whole

u/MysteriousPepper8908 • 3 points • 2026-01-16 05:47 UTC

I would agree with this. On a macro scale, AI isn't doing any harm that Youtube and Facebook weren't doing much faster but there's no reason we need to be plopping down data centers in the middle of neighborhoods as they can have nasty localized effects and you should probably do what you can to resist it if someone tries to build a data center in your neighborhood.

u/CuirPig • 1 points • 2026-01-25 07:10 UTC

From my understanding, datacenters require both Power Stations and a lot of space for the infrastructure. It seems to me that they would not be able to build in neighborhoods. They might build next to a neighborhood, providing lots of jobs, investing money in the local community, sponsoring education for the area, etc.

Microsoft, for example, has agreed to pay well over the going rate for utilities for their next datacenter. That money will help the local community.

u/shortandpainful • 3 points • 2026-01-16 05:40 UTC

Hank Green is fantastic. I recently found out he's John Green's brother (before that he was just a guy I saw on YouTube/Twitter sometimes being based).

Another good source for this is the podcast Science Vs, which did a recent episode on this.

u/Glass-Ad672 • 2 points • 2026-01-16 05:34 UTC

thanks